Mod-4 obstructions for completing the rank-47 $\mathbb{F}_2$ algorithms for 4×4 matrix multiplication

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Abstract

The best known bilinear algorithms for multiplying two 4×4 matrices use 47 multiplications over fields of characteristic two (AlphaTensor, 2022; Kauers–Moosbauer, 2023) and 48 multiplications over rings containing $\frac{1}{2}$ (AlphaEvolve, 2025; Dumas–Pernet–Sedoglavic, 2025), while every known algorithm that is valid over all rings, for instance over $\mathbb{Z}/2^{64}\mathbb{Z}$, uses 49. It is known that the two rank-47 schemes over $\mathbb{F}_2$ do not lift to characteristic zero. We show that they cannot be completed to rank-48 schemes over $\mathbb{Z}/4\mathbb{Z}$ either: there is no decomposition of the $4 \times 4 \times 4$ matrix multiplication tensor over $\mathbb{Z}/4\mathbb{Z}$ (hence none over $\mathbb{Z}$ or over the 2-adic integers) consisting of 48 rank-one terms of which 47 reduce modulo 2 to the AlphaTensor scheme and the remaining one vanishes modulo 2. More generally, no correction of the form $2\,x \otimes M$, with an arbitrary 16×16 block M, can repair the mod-4 defect of the lifted scheme. The proof reduces the question to linear algebra over $\mathbb{F}_2$ and is certified by 1585 explicit linear functionals. The reduction is formalised in Lean 4 with Mathlib; the finite checks on the explicit certificate are run with `native_decide`; the link between the data and the matrix multiplication tensor is kernel-checked. The same statement holds, with its own certificate and the same formalisation, for the Kauers–Moosbauer scheme. Both schemes are rigid in the flip graph: an exhaustive exploration through rank 48 (flips, plus transitions, reductions) reaches no other scheme of rank ≤ 47. We also report the analogous computation for the rank-73 $\mathbb{F}_2$ scheme of $\langle 4, 5, 5\rangle$ and 2-adic integrality measurements for several rational records.

1 Introduction

A bilinear algorithm (a *scheme*) of rank r for the product of an $n \times m$ matrix A and an $m \times p$ matrix B over a commutative ring K consists of vectors $u_t \in K^{nm}$, $v_t \in K^{mp}$, $w_t \in K^{np}$ $(1 \leq t \leq r)$ with

$$\sum_{t=1}^{r} u_t \otimes v_t \otimes w_t \;=\; \mathcal{T}_{n,m,p}, \qquad (\mathcal{T}_{n,m,p})_{(i,j),(j',k),(i',k')} = \delta_{ii'}\delta_{jj'}\delta_{kk'}. \tag{1.1}$$

Equivalently $C_{ik} = \sum_t w_t[i,k]\,(\sum u_t[i',j]A_{i'j})(\sum v_t[j',k']B_{j'k'})$, also for non-commuting matrix entries, so a scheme can be applied recursively. Over $K = \mathbb{Z}$ a scheme is valid in every ring.

For 4×4 matrices, Strassen's algorithm applied twice gives 49 multiplications over $\mathbb{Z}$. AlphaTensor [1] found a scheme of rank 47 over $\mathbb{F}_2$, and Kauers and Moosbauer [2] found another one with flip graphs. AlphaEvolve [3] found rank 48 over $\mathbb{C}$, and Dumas, Pernet and Sedoglavic [4, 5, 6] turned it into rational schemes; all of these use the coefficient $\frac{1}{2}$. Whether rank 48 is possible

over $\mathbb{Z}$ (equivalently, over every ring) is open; it matters for exact integer arithmetic modulo 2^{64}, where division by 2 is not available. The rank-47 $\mathbb{F}_2$ schemes do not lift: their first Hensel step is obstructed, and in fact they lift to no finite extension of $\mathbb{Z}_2$ [7].

A natural way around this is to allow *defect terms*: a 2-adic scheme of rank 48 in which one term is divisible by 2 reduces modulo 2 to a scheme of rank 47. So one may ask whether one of the known $\mathbb{F}_2$ schemes of rank 47 is the reduction of such a scheme. We show that it is not, already modulo 4.

Main result. Let $S = (u_t, v_t, w_t)_{t\le 47}$ be the AlphaTensor $\mathbb{F}_2$ scheme for $\mathcal{T} = \mathcal{T}_{4,4,4}$ (data in the Lean file).

Theorem 1.1. *There are no $a_t, b_t, c_t \in (\mathbb{Z}/4)^{16}$ with $a_t \equiv u_t$, $b_t \equiv v_t$, $c_t \equiv w_t \pmod 2$ ($t = 1, \dots, 47$), $x \in (\mathbb{Z}/4)^{16}$ and $M \in (\mathbb{Z}/4)^{16\times 16}$ such that*

$$\sum_{t=1}^{47} a_t \otimes b_t \otimes c_t \;+\; 2\,(x \otimes M) \;=\; \mathcal{T} \quad \text{over } \mathbb{Z}/4.$$

Corollary 1.2. *There is no scheme of rank 48 for 4×4 matrix multiplication over $\mathbb{Z}/4$, over $\mathbb{Z}$, over $\mathbb{Z}_2$, or over any subring of $\mathbb{Q}$ in which 2 is not invertible, whose reduction modulo 2 consists of the 47 terms of S (in some order) and one term that is 0 modulo 2. The same holds for every image of S under the symmetry group of $\mathcal{T}$ over $\mathbb{F}_2$.*

Proof. Reduce such a scheme modulo 4. A rank-one term $\alpha\otimes\beta\otimes\gamma$ that vanishes modulo 2 has a factor divisible by 2, so it equals $2\,x\otimes M$ with M the outer product of the other two factors (after moving the 2 to the first factor). Over $\mathbb{F}_2$ a nonzero rank-one tensor determines its factors, so the other terms reduce factorwise to the terms of S. This contradicts Theorem 1.1. For an image $g\cdot S$, $g = (P, Q, R) \in GL_4(\mathbb{F}_2)^3$ (or a cyclic/transposition symmetry), lift g to $\tilde g$ over $\mathbb{Z}/4$ (determinants are odd); $\tilde g^{-1}$ maps a completion of $g\cdot S$ to one of S, because $\mathcal{T}$ is invariant and divisibility by 2 is preserved. □

Theorem 1.1 also excludes any number of extra 2-divisible terms that share their first factor. By a further computation (not formalised) the same holds for extra terms sharing their second or their third factor; all formal statements also hold for the Kauers–Moosbauer scheme (Section 4).

2 Reduction to linear algebra over $\mathbb{F}_2$

Write $\bar u_t \in \{0,1\}^{16}$ also for the integer lift of u_t, and define the integer tensor

$$\sum_t u_t \otimes v_t \otimes w_t = \mathcal{T} + 2R + 4(*), \qquad R \in \{0,1\}^{16\times 16\times 16},$$

which exists because S is a scheme over $\mathbb{F}_2$. Let $J : \mathbb{F}_2^{3\cdot 47\cdot 16} \to \mathbb{F}_2^{4096}$ be the derivative of the Brent map at S, $J(\alpha,\beta,\gamma) = \sum_t \alpha_t\otimes v_t\otimes w_t + u_t\otimes\beta_t\otimes w_t + u_t\otimes v_t\otimes\gamma_t$.

Lemma 2.1. *If a, b, c, x, M are as in Theorem 1.1, write $a_t = u_t + 2\alpha_t$, $b_t = v_t + 2\beta_t$, $c_t = w_t + 2\gamma_t$. Then, over $\mathbb{F}_2$,*

$$R + J(\bar\alpha, \bar\beta, \bar\gamma) + \bar x \otimes \bar M = 0.$$

Proof. $(u+2\alpha)(v+2\beta)(w+2\gamma) = uvw + 2(\alpha vw + u\beta w + uv\gamma)$ in $\mathbb{Z}/4$. Summing over t and using the hypothesis, $2\,(R + J(\alpha,\beta,\gamma) + x\otimes M) = 0$ in $\mathbb{Z}/4$, i.e. the bracket is even. □

Lemma 2.2 (witness)**.** *Let $\ell : \mathbb{F}_2^{4096} \to \mathbb{F}_2$ be linear with (i) $\ell(R) = 1$, (ii) $\ell \circ J = 0$, and let $x \in \mathbb{F}_2^{16}$ satisfy (iii) $\sum_p x_p\, \ell(e_p \otimes e_q \otimes e_s) = 0$ for all q, s. Then $R + J(d) + x \otimes M \neq 0$ for all d and all M.*

Proof. Apply ℓ: $\ell(R) + \ell(J(d)) + \sum_{q,s} M_{qs} \sum_p x_p \ell(e_p \otimes e_q \otimes e_s) = 1 + 0 + 0$. □

Condition (iii) says that x lies in the left kernel of the 16×256 matrix of ℓ. A *certificate* is a family of functionals $\ell_1, \dots, \ell_N$ satisfying (i), (ii), together with subspaces L_i of their left kernels whose union is $\mathbb{F}_2^{16}$. By Theorems 2.1 and 2.2 a certificate proves Theorem 1.1.

Finding certificates. The cokernel of J has dimension $4096 - \operatorname{rank} J = 4096 - 2117 = 1979$ (the kernel of J has dimension 139, in agreement with [7], where it is identified with the tangent space of the symmetry action). For a single $x \neq 0$ the obstruction $R \notin \operatorname{im} J + x \otimes \mathbb{F}_2^{256}$ is a linear algebra test; we first ran it for all 65535 vectors x (a C program, about 10 seconds), and found no x passing. To produce a compact certificate we grow subspaces L greedily: a functional with (i), (ii) and $L \otimes \mathbb{F}_2^{256} \subseteq \ker \ell$ exists as long as $256 \dim L$ is small compared with 1979, and in practice up to $\dim L = 7$. A greedy cover of $\mathbb{F}_2^{16} \setminus 0$ by 1585 such 7-dimensional subspaces gives the certificate used in Lean (1581 for the Kauers–Moosbauer scheme).

3 Formal verification

The Lean file (`lean/Main.lean`, Lean 4.35.0 with Mathlib) contains

- a generic theorem `MM4.no_Z4_completion_of_facts`: for *arbitrary* data u, v, w, R, functionals ℓ_i, subspace bases and a selector, the finite facts F0 ($\sum_t uvw = \mathcal{T} + 2R$ over $\mathbb{Z}/4$), F1–F4 (conditions (i), (ii) for every ℓ_i), F5a (basis vectors lie in the left kernels) and F5b (every $x \in \mathbb{F}_2^{16}$ is a combination of the basis of the selected witness) imply the statement of Theorem 1.1. Its proof (Theorems 2.1 and 2.2) is a complete Lean proof using only Mathlib;
- the explicit data of S, of R and of the certificate, and proofs of F0–F5b; and
- the main theorem

```
theorem AT47.no_Z4_completion :
  Not (exists (a b c : Fin 47 -> Fin 16 -> ZMod 4) (x : Fin 16 -> ZMod 4)
        (M : Fin 16 -> Fin 16 -> ZMod 4),
    (forall t s, (a t s).val % 2 = u t s) /\
    (forall t s, (b t s).val % 2 = v t s) /\
    (forall t s, (c t s).val % 2 = w t s) /\
    forall p q s : Fin 16,
      (Finset.univ.sum fun t : Fin 47 => a t p * b t q * c t s)
        + 2 * (x p * M q s) = MM4.T4 p q s)
```

(written here in ASCII notation; the Lean file uses the usual Unicode symbols.) and the corollary `AT47.no_rank48`: there are no a, b, c : `Fin 48` $\to (\mathbb{Z}/4)^{16}$ whose first 47 terms reduce modulo 2 to S, whose last term has a factor $\equiv 0 \pmod 2$, and with $\sum_{t<48} a_t \otimes b_t \otimes c_t = \mathcal{T}$ over $\mathbb{Z}/4$ (deduced from the main theorem by the generic lemma `MM4.no_rank_succ_of_no_completion`). Here `MM4.T4 p q s` is 1 if $\lfloor p/4 \rfloor = \lfloor s/4 \rfloor$, $p \bmod 4 = \lfloor q/4 \rfloor$, $q \bmod 4 = s \bmod 4$ and 0 otherwise (index $4i + j$ for entry (i, j)).

Statement audit. `u t s` is the s-th bit of the explicit t-th coefficient mask of the AlphaTensor scheme, read from `factorizations_f2.npz` in the AlphaTensor repository and converted to our index convention (the C factor is transposed there); F0 shows that (u, v, w) is indeed a scheme for $\mathcal{T}$ modulo 2. The hypothesis `(a t s).val % 2 = u t s` says exactly that $a_t \equiv u_t \pmod 2$. The conclusion is the negation of the existence of a mod-4 decomposition of the stated shape.
Axioms. `#print axioms` reports `propext`, `Classical.choice`, `Quot.sound` and one auxiliary axiom `..._native.native_decide.ax_1_1` for each of the six certificate facts F1, F2, F3, F4, F5a, F5b, which are evaluated with `native_decide` (so the compiled evaluation of these Boolean computations is trusted); the fact F0, which ties the data to the matrix multiplication tensor, is checked by the kernel (`decide +kernel`). See `lean/lean_output.txt` (compilation about 31 minutes and 13 GB of memory on one core; `lean/lean_output_KM47.txt` for the second scheme). To keep the computation small, the sums in F2–F5a run only over the supports of the (0/1) coefficient vectors; the equality with the full sums used in the generic theorem is proved in Lean. The certificate facts were also verified by an independent Python script (`verify_cert.py`).

4 Other schemes and related computations

Kauers–Moosbauer 47. The scheme `444-47-mod2` of the flip-graph repository [2] has the same Jacobian rank 2117; no x passes the test in any of the three modes. A certificate of 1581 functionals proves the analogue of Theorem 1.1 and of Theorem 1.2 in Lean (`KM47.no_Z4_completion`, `KM47.no_rank48`, file `lean/MainKM47.lean`, same axioms).

Rigidity of the two schemes. The two schemes are not equivalent under the symmetry group of $\mathcal{T}$: the multisets of triples $(\operatorname{rank} u_t, \operatorname{rank} v_t, \operatorname{rank} w_t)$ of the coefficient matrices differ, also after permuting the three modes (for instance, the numbers of factors of rank $1, 2, 3, 4$ in the three modes are $(20, 12, 15, 0)$, $(20, 12, 15, 0)$, $(9, 24, 13, 1)$ for AlphaTensor's scheme and $(20, 12, 15, 0)$, $(11, 24, 11, 1)$, $(19, 12, 16, 0)$ for the Kauers–Moosbauer scheme). Neither scheme admits a flip: no two of its terms share a factor, so each is an isolated vertex of the rank-47 flip graph over $\mathbb{F}_2$. In fact rank 48 does not help either: we applied each of the $47 \cdot 46 \cdot 3 = 6486$ split ("plus") transitions to rank 48, explored the complete rank-48 flip component of each result (971 components with 37,848 schemes in total for AlphaTensor's scheme, 981 components with 34,209 schemes for the Kauers–Moosbauer scheme) and applied every possible reduction to every scheme met; the only scheme of rank at most 47 obtained is the starting one. Hence other rank-47 schemes can be reached from these two only through rank 49 or higher. (As a sanity check, the same program finds 143 further rank-23 schemes around a rank-23 $\mathbb{F}_2$ scheme for $\langle 3, 3, 3\rangle$ after exploring about 42,000 rank-24 schemes; there a single rank-24 flip component already has more than 27,000 elements.) None of the $37{,}848 + 34{,}209$ rank-48 $\mathbb{F}_2$ schemes met in this exploration lifts to a rank-48 decomposition over $\mathbb{Z}/4$: for each of them the defect $(\mathcal{T} - \sum_t u_t v_t w_t)/2 \bmod 2$ is not in the image of its Jacobian (the same test confirms that Strassen's algorithm squared, reduced modulo 2, does lift). Going one level up, we sampled 384 (AlphaTensor) and 488 (Kauers–Moosbauer) random splits of rank-48 schemes of these neighbourhoods to rank 49 and explored their rank-49 flip components by breadth-first search (up to 10^5 schemes each; about $2.2 \cdot 10^7$ rank-49 schemes per starting scheme, counted with repetition; about five core-hours): every reduction of a scheme met leads back into the explored rank-48 neighbourhood (or to a trivial split of the starting scheme), so neither a further rank-47 scheme nor a new rank-48 flip component was found. Random flip-graph walks through ranks ≤ 50 (`find_alternative_schemes` of [10], about five core-hours) did not produce another rank-47 scheme either. These searches are not exhaustive.

$\langle 4,5,5\rangle$**.** The best rank over $\mathbb{Q}$ is 76, while Arai, Ichikawa and Hukushima found an $\mathbb{F}_2$ scheme of rank 73 [8] (rank-74/75 $\mathbb{F}_2$ schemes do not lift [9]). For this rank-73 scheme (rank $J = 4527$, cokernel 5473) an exhaustive scan of all $2^{20} - 1$ first factors shows that no completion to rank 74 over $\mathbb{Z}/4$ by one 2-divisible term exists. (Computational, C program validated on planted instances; no certificate file.)

2-adic integrality of rational records. For a rational scheme with 2-power denominators we searched the product of the three Bruhat–Tits buildings of GL_n, GL_m, GL_p over $\mathbb{Q}_2$ for a change of basis making all terms integral. The best total deficiency found (0 would give an integral scheme of the same rank) is 32 for the Dumas–Pernet–Sedoglavic 48 (sixteen terms keep a factor $\frac{1}{4}$, also for all 552 rational rank-48 schemes of [7]), 24 for the AlphaEvolve $\langle 2,4,5\rangle$:32, 30 for $\langle 3,4,7\rangle$:63 and 88 for Smirnov's $\langle 3,3,6\rangle$:40 (all terms non-integral). These are search results, not lower bounds.

5 Limitations and open questions

- The theorem concerns the two known rank-47 $\mathbb{F}_2$ schemes (and their symmetry orbits). Other $\mathbb{F}_2$ schemes of rank 47 (none is reachable from the two known ones through rank ≤ 48), rank-48 schemes over $\mathbb{F}_2$ that lift modulo 4 (none among the 72,057 near the two schemes), schemes with two or more defect terms with different factors, and lifts over ramified extensions of $\mathbb{Z}_2$ are not excluded.
- Whether $\langle 4,4,4\rangle$ has rank 48 over $\mathbb{Z}$ remains open, as do $\langle 4,5,5\rangle$ at rank 75 and Kauers and Moosbauer's question whether 42 is optimal for $\langle 3,3,6\rangle$ in characteristic two.
- The finite facts are checked with `native_decide`, which trusts the Lean compiler; the algebraic reduction is kernel-checked.

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